

GAUTAM CHOUHAN

Indore, India

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EDUCATION

Indian Institute of Information Technology, Sri City

B.Tech in Computer Science and Engineering

Aug. 2023 – May 2027

CGPA: 8.95 / 10

EXPERIENCE

Full Stack Engineer Intern

July 2025 – October 2025

Soundverse AI

Remote

- Redesigned the AI generation flow by introducing a **Redis-based fetch mechanism**, reducing project creation **latency by 80%** and significantly improving overall user experience.
- Collaborated closely with backend engineers to integrate and ship new **AI-powered tools**, ensuring seamless end-to-end functionality and production readiness.
- Integrated **Veena DAW** support via **iFrame integration and JWT-based authentication**.
- Refactored the **account deletion flow**, improving security, handling edge cases more gracefully & reducing user-facing errors.
- Integrated new **model versions** and resolved critical **production issues**, improving platform stability across core user workflows.

PROJECTS

Resonance AI | [🔗](#) | *Next.js 16, React 19, TypeScript, tRPC, Prisma, PostgreSQL, Modal, Python, Clerk*

- Developed an AI Text-to-Speech platform for high-quality, expressive voice synthesis.
- Built a scalable inference backend using Python and Modal (Chatterbox TTS) for on-demand GPU compute.
- Ensured end-to-end type safety across the stack using tRPC, TypeScript, and Zod.
- Implemented multi-tenant organization management with Clerk and distributed audio storage with Cloudflare R2.
- Enabled real-time browser-based recording for custom voice profiling and management.

goShell | [🔗](#) | *Go, Concurrency*

- Built a Unix-like shell in Go with a modular architecture separating parsing and command execution.
- Implemented pipelines (`|`), background jobs (`&`), and file redirection (`>`) using goroutines and process management.
- Developed an interactive terminal with command history, tab completion, and built-in commands (`cd`, `pwd`, `jobs`, `exit`).
- Managed shell state, environment variables, and process lifecycle for reliable command execution.

Kinetic | [🔗](#) | *Next.js, FastAPI, Python, JavaScript, Docker, Tailscale*

- Engineered a peer-to-peer GPU sharing platform with real-time provider matching based on live hardware capacity.
- Built the FastAPI control plane for matchmaking, heartbeat ingestion, and rental lifecycle management.
- Designed the host agent to report GPU, CPU, RAM, and temperature telemetry while enforcing container-level resource limits.
- Implemented secure private connectivity over Tailscale for host provisioning and remote machine control.

TECHNICAL SKILLS

Languages: Go, JavaScript, TypeScript, Java, C, Python

Frontend Development: React.js, Next.js, HTML, CSS, TailwindCSS

Backend Development: Node.js, Express.js, REST APIs, FastAPI

Database Management: MongoDB, PostgreSQL, Prisma (ORM)

Tools & Technologies: Git, AWS, VS Code

POSITIONS OF RESPONSIBILITY

Domain Lead (Problem Setter) | *Gradient Programming Club*

Curated algorithmic problems and organized multiple coding contests for college students.

Co-Lead of Tech Team | *College Annual Fests (2025–26)*

Led the developer team to build official web platforms for flagship fests **Abhisarga** and **Utkrishta**.

Tech Team Member | *Web3SSH*

Contributed to technical execution for a student-driven summer school program focused on Web3.